ChromaDB Search & Filter Exercises

Metadata Filtering · Full Text Search · Combined Queries

4 exercises · ~40 minutes · individual or pairs

All four exercises share one starter file and one knowledge base. You only need to add code in the marked sections — do not modify the setup above the exercise markers.

The Knowledge Base

The collection contains 18 documents from a fictional university IT helpdesk, split across six categories. Each document has three metadata fields:

Field	Type	Values
`category`	string	`"vpn", "email", "software", "network", "accounts", "printing"`
`priority`	string	`"high", "medium", "low"`
`year`	integer	`2024 or 2025`
`verified`	boolean	`True or False`

Quick Reference

collection.get() — direct fetch, no similarity ranking

```
collection.get(where={"field": "value"})                    # exact match
collection.get(where={"field": {"$gt": 5}})                 # numeric comparison
collection.get(where={"$and": [{...}, {...}]})                # combine filters
collection.get(where_document={"$contains": "word"})          # text contains
collection.get(where_document={"$not_contains": "word"})      # text does not contain
```

collection.query() — semantic similarity search + optional filters

```
collection.query(
    query_texts=["your question here"],
    n_results=3,
    where={"field": "value"},               # optional metadata filter
    where_document={"$contains": "word"},   # optional text filter
    include=["documents", "metadatas", "distances"]
)
```

Exercise 1 — Basic Metadata Filter

Concept: using where with a simple equality filter on collection.get()

Task

Use collection.get() with a where filter to retrieve all documents where category equals "vpn".

- Use the where parameter with a simple equality expression
- Pass include=["documents", "metadatas"] to see both text and metadata
- Pass your results to print_results() to display them

Expected result

3 documents — doc-000, doc-001, doc-002

Write your code here

```
result = collection.get(
    # your where filter here
    include=["documents", "metadatas"]
)
print_results("Exercise 1", result)
```

Observations (write in)

Exercise 2 — Combined Metadata Filters

> Concept: combining multiple metadata conditions with $and and $or logical operators

Task A

Use collection.get() to retrieve all documents where priority is "high" AND year is 2025 AND verified is True.

- Use $and to combine three conditions
- For the boolean field, use {"verified": True} — not a string

Expected result

7 documents spanning vpn, email, network, and accounts categories

Task B — Extension

Use collection.get() to retrieve all documents where category is "software" OR category is "printing".

- Use $or to match either category
- You should get 6 documents

Write your code here

```
# Task A
result_a = collection.get(
    # your $and filter here
    include=["documents", "metadatas"]
)
print_results("Exercise 2A", result_a)

# Task B
result_b = collection.get(
    # your $or filter here
    include=["documents", "metadatas"]
)
print_results("Exercise 2B", result_b)
```

Observations (write in)

Exercise 3 — Full Text Search

Concept: filtering by document content with where_document using $contains and $not_contains

Task A

Use collection.get() with where_document to retrieve all documents whose text contains the word "student".

- Use where_document={"$contains": "student"}
- Note: $contains is a substring match, not semantic — it looks for the exact string

Task B — Extension

Narrow the results: find documents that contain "student" but do NOT contain "password".

- Combine $contains and $not_contains inside a $and operator in where_document
- Compare the result count to Task A — how many documents were excluded?

Write your code here

```
# Task A
result_a = collection.get(
    # your where_document filter here
    include=["documents", "metadatas"]
)
print_results("Exercise 3A", result_a)

# Task B
result_b = collection.get(
    # your combined where_document filter here
    include=["documents", "metadatas"]
)
print_results("Exercise 3B", result_b)
```

Observations (write in)

Exercise 4 — Semantic Query + Combined Filters

Concept: combining collection.query() (semantic search) with both where (metadata) and where_document (text) filters

Task

Use collection.query() to find documents semantically related to "how do I print documents on campus". Restrict results to:

- Only documents from category "printing" — use where
- Only documents whose text contains the word "page" — use where_document
- Request n_results=5 and include distances in the output

Reflection questions

- How many results are returned? Why might it be fewer than n_results=5?
- Look at the distance values. Are they close to 0 or far from 0? What does that tell you?
- Remove the where filter and run again. How does the result set change?

Write your code here

```
result = collection.query(
    query_texts=["how do I print documents on campus"],
```

```
    n_results=5,
    # add your where filter here
    # add your where_document filter here
    include=["documents", "metadatas", "distances"]
)
print_results("Exercise 4", result, show_distances=True)
```

Observations (write in)

Bonus challenges

1. Use a $gte numeric filter: find all documents from year 2025 or later.
2. Use $in: find all documents whose priority is either "high" or "medium".
3. Combine metadata and text filters: find high-priority documents that contain the word "MFA".
4. Use collection.get(include=[]) to retrieve only document IDs — no text, no metadata. When is this useful?